Humanoid Weekly Update

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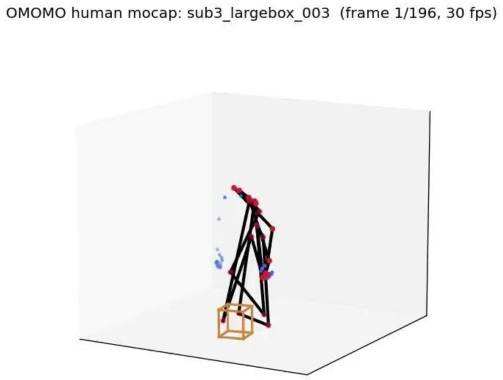
Problem we are trying to solve:

- **How can robots detect, reason about, and recover from failures in their perception system during real-world operation?**

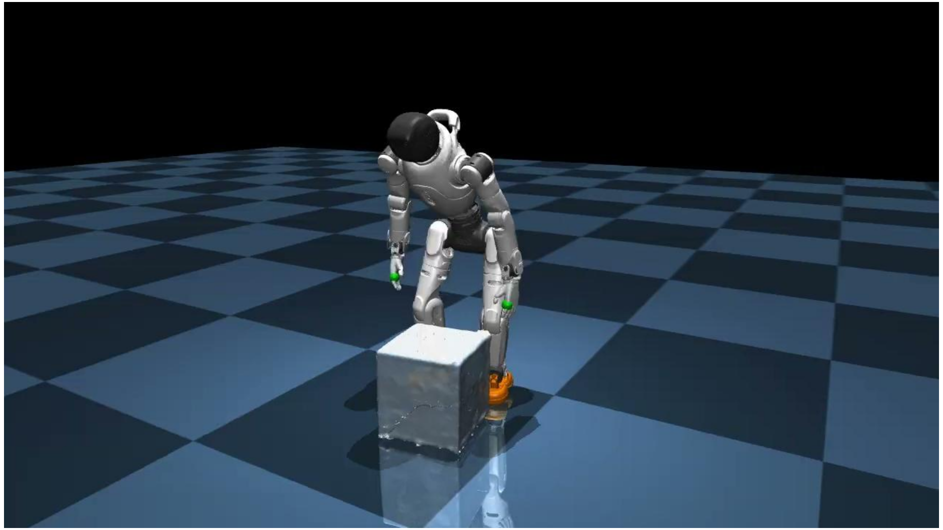
- **Focus of this week was to train policies to use in the future when reasoning**

Reference Motion: Human MoCap → Retargeted X2

Source — human SMPL-H MoCap



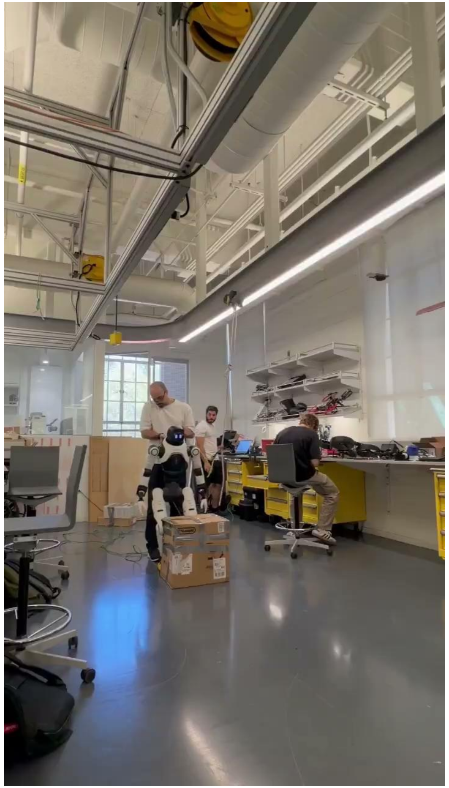
Retargeted — AgiBot X2 (31-DOF)



NEW POLICY

Box Pickup Tests

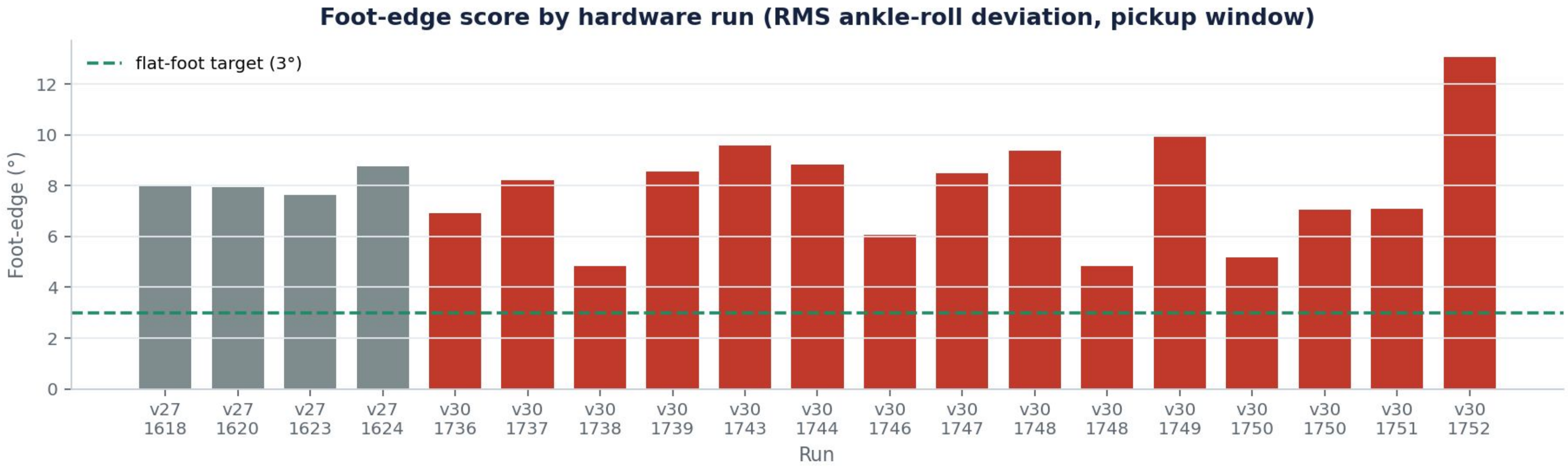
Previous Policy



V2

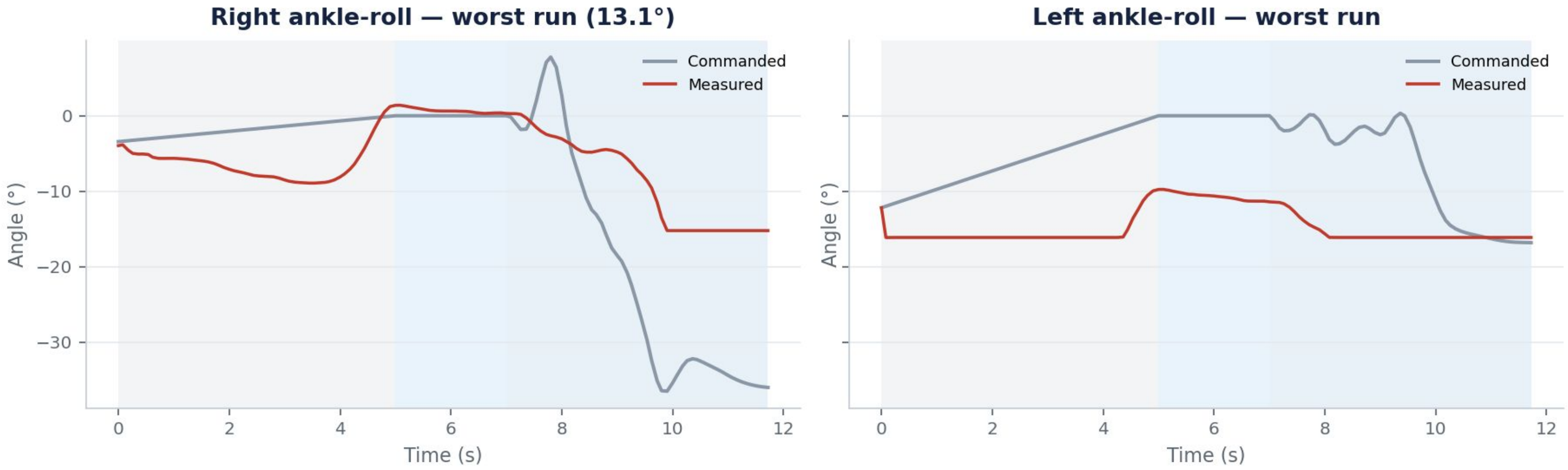
V3

Foot-edge severity across all runs



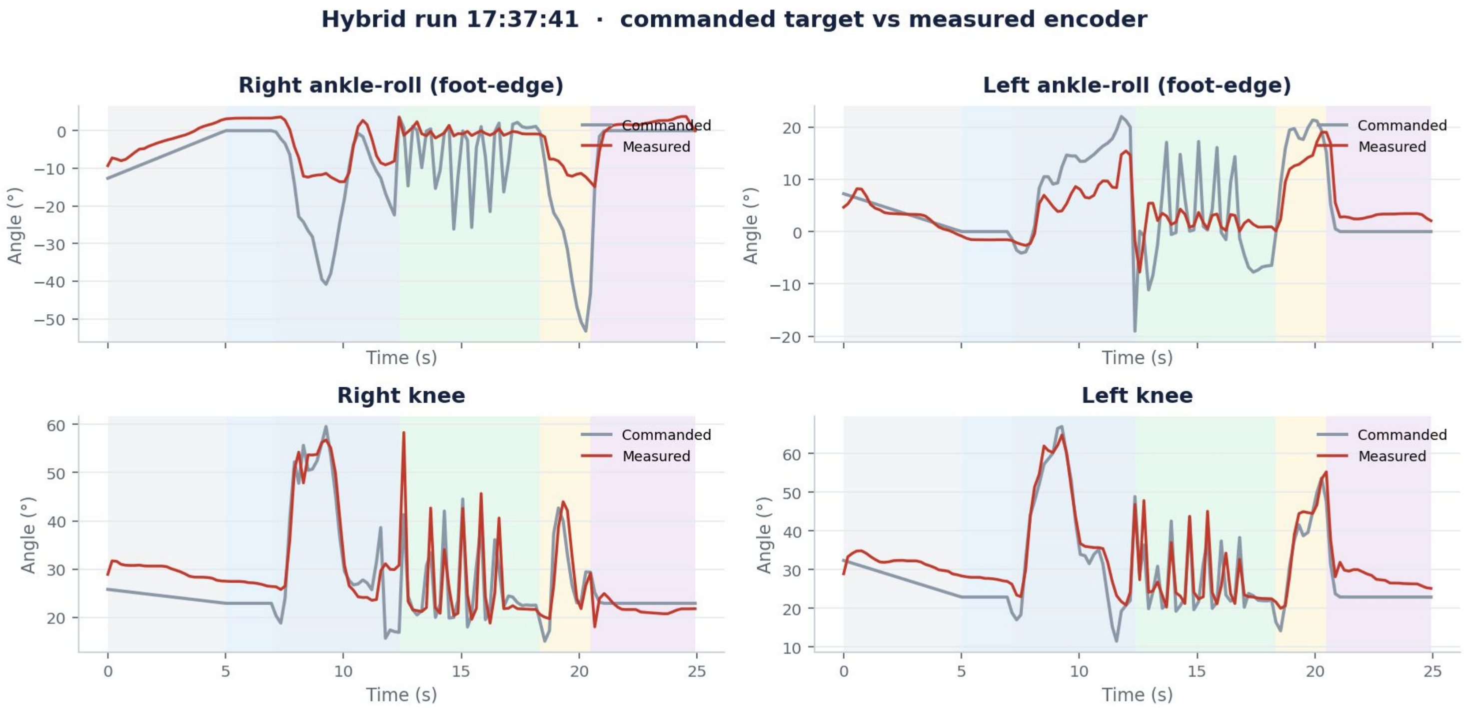
RMS |ankle-roll| during pickup. Red = v30, gray = v27. Dashed = 3° flat-foot target.

Worst run — edge-standing signature



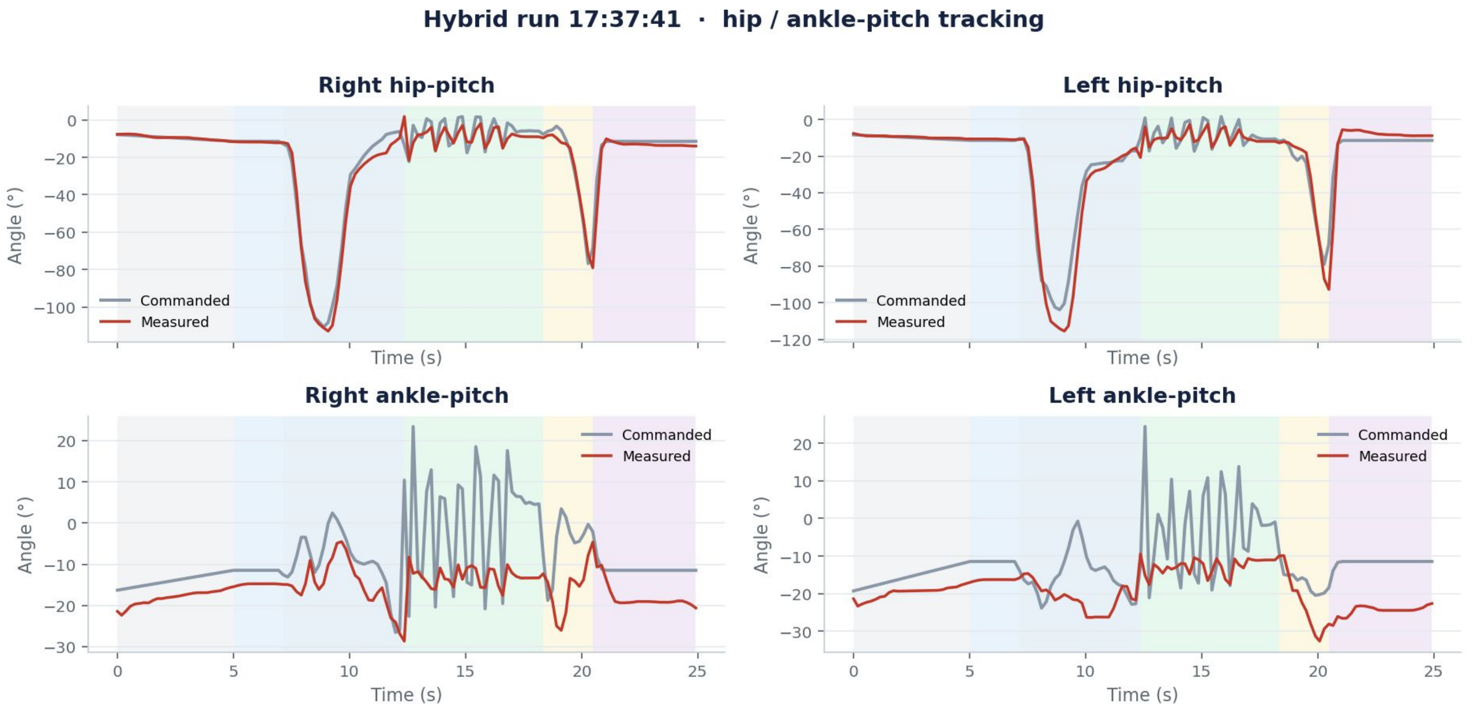
Run 17:52:22. Commanded ankle-roll nearly flat; measured swings hard onto the edge.

Desired vs actual — ankles and knees



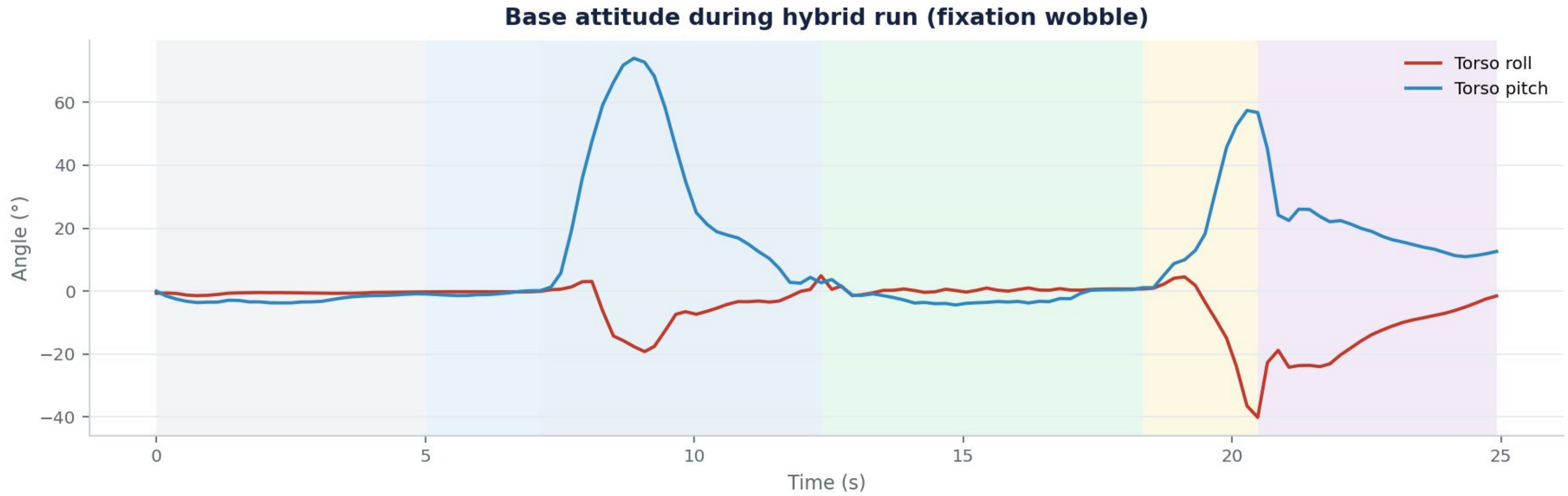
Hybrid run 17:37:41. Shaded bands = ramp / settle / pickup / carry / setdown / done.

Desired vs actual — hips and ankle-pitch



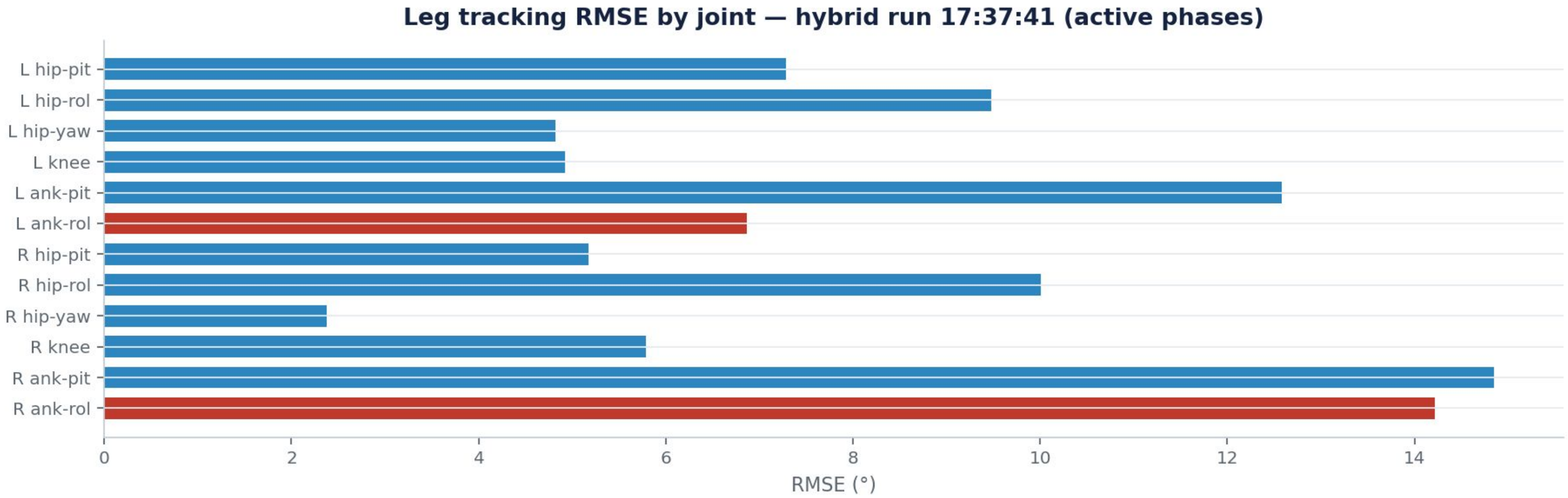
Same hybrid run. Hip tracking holds better than ankle-roll — failure is concentrated at the foot.

Base attitude — fixation wobble



Repeated roll swings during pickup = rebalancing on foot edges, not a clean plant.

Per-joint tracking error (legs)



Ankle-roll joints (red) dominate leg RMSE — matches the edge-standing observation.

Why prior rewards missed the failure

- `foot_slip`: zero when the foot is tilted but stationary — edge-stand is free.
- `feet_contact_loss`: edge still registers force in PhysX, so it counts as planted.
- `feet_anchor` / slip: address XY drift and sliding, not foot roll about the edge.
- No term measured ankle-roll link tilt → foot-tilt bounced 4°–29° across checkpoints with no trend.

Penalties added to fix hardware failures

Term	W	What it fixes
foot_slip	−3	Skating under load (sim → thrash on HW)
feet_contact_loss	−2	Stepping loophole (lift to evade slip)
feet_anchor	−2→−1	Stance widen / permanent XY drift
foot_not_flat	−3	Edge-standing (sin of foot tilt, contact-gated)
feet_edge_contact	−2	Edge counts as not planted ($\theta \gtrsim 10°$)
ori tracking	×2	Left torso lean during pickup
end-hold in motion	—	Fall after set-down (3 s upright practice)

New foot terms (v31)

Foot not flat — smooth tilt cost while loaded

$$r_{\text{tilt}} = \sum_f (w_r|g_f^y| + w_p|g_f^x|)\,\mathbf{1}_{\text{contact}}$$

Edge contact — hard count of edge-planted feet

$$r_{\text{edge}} = \sum_f \mathbf{1}[\text{contact} \wedge \sin\theta_f > \theta_{\text{th}}]$$

Contact loss — airborne foot (stepping)

$$r_{\text{air}} = \sum_f \mathbf{1}[|F_f| < \tau]$$

Anchor — XY drift from reset plant

$$r_{\text{anchor}} = \sum_f |p_f^{xy} - p_{f,0}^{xy}|$$

Mapping: symptom → penalty

- Edge-standing / oscillating on feet → `foot_not_flat` (−3) + `feet_edge_contact` (−2).
- Stepping during pickup/hold/set-down → `feet_contact_loss` (−2) + stronger `foot_slip` (−3).
- Stance growing apart → `feet_anchor` (drift from plant pose).
- Torso roll left on lift → double `motion_global_ref_orientation` weight.
- Fall after drop → bake 3 s end-hold into reference motion variants.

Deployment notes from the data

- Hybrid deploy freezes the WBT clock at hold_frame (~261) for the walk splice — looks like a mid-hold stop.
- Full continuous pickup→set-down uses `deploy_x2_box_pickup.py`, not the hybrid script.
- Always pass `--box-policy` explicitly; default path previously loaded stale v27.
- Judge new checkpoints by foot-tilt + post-set-down upright, not mean reward alone.

Takeaways

- Hardware logs localize the failure to ankle-roll / foot tilt, not a generic tracking miss.
- Sim contact force ≠ flat sole — need an explicit flatness / edge objective.
- Planted-feet suite + foot-flat terms close the loopholes that produced IRL edge-standing and steps.
- Hold and set-down need dedicated sampling; short episodes never reach those phases.

Thank you!

NEW POLICY

Policy Training — RL Tracking

4096	**50 Hz**	**31**	**PPO**
parallel envs	control (200 Hz sim)	actuated DOF	λ=0.95 · γ=0.99

- **Framework:** IsaacLab (Isaac Sim) + holosoma.
- **Actor:** blind MLP 512–256–128 (ELU) — prev action, base ang-vel, dof pos/vel, motion phase, ref orientation.
- **Critic:** privileged obs adds base lin-vel + object & body state.
- **Key reward:** two-hand-to-box proximity as a product term — both palms must reach the box simultaneously.
- **Status:** currently training; deploy on physical X2 after the run.

Actor (blind) → 512–256–128 → 31 DOF

Critic (privileged: + object & body state)

Two-hand grasp reward

$$r_{hands} = \exp\left(-\frac{\sum_{i\in\{L,R\}} \|p_i^{hand} - p^{box}\|^2}{\sigma^2}\right), \quad \sigma = 0.25\,m$$

Product (not mean) over hands forces bimanual contact.

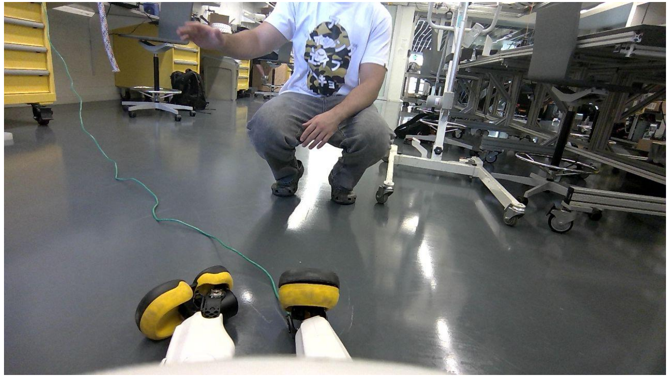
VLM Perception — Setup

- **Goal:** scene understanding from the X2 head RGB camera; frames streamed to a workstation for evaluation.
- **Three VLMs served locally via Ollama:** Qwen2.5-VL 7B, Qwen2.5-VL 3B, Moondream (~1.8B).
- **Protocol:** 15 s clip @ ~20 fps → 12 evenly-spaced frames; identical prompt, temperature 0.2; one model resident at a time (8 GB VRAM).

Prompt (identical for all models)

“You are a robot's vision system. List the main objects and people you see and roughly where they are (left/center/right). Then one short summary sentence. Be concise.”

VLM Qualitative Comparison — Frame 224 (t = 11.3 s)



Qwen2.5-VL 7B (2.80 s | 51.6 tok/s)

Objects: yellow toolbox, green cord, wheels, white t-shirt with text, gray pants. People: man in white t-shirt squatting. Summary: a person interacting with a wheeled device in an industrial setting.

Qwen2.5-VL 3B (2.60 s | 107.9 tok/s)

A person squatting in the center of the room; yellow and black robotic feet on the floor, likely attached to a robot or machine.

Moondream (0.67 s | 203.4 tok/s)

Person in white shirt, gray pants, black shoes, and yellow shoes standing near a desk with a green cord on it.

VLM Performance Metrics

Model	Params	Mean lat.	Median	Rate	Gen speed
Qwen2.5-VL 7B	7 B	2.87 s	2.89 s	0.35 Hz	51.3 tok/s
Qwen2.5-VL 3B	3 B	2.45 s	2.50 s	0.41 Hz	107.6 tok/s
Moondream	~1.8 B	0.61 s	0.62 s	**1.65 Hz**	**215.4 tok/s**

- Warm, end-to-end latencies on an RTX 5070 (8 GB); 12 frames, single 15 s clip.
- Moondream is ~4–5× faster — the only model near real-time (~1.7 Hz).
- Qwen2.5-VL 7B usually gives the most accurate scene descriptions; 3B is slower than Moondream yet less accurate on this scene.

Older slides below

Box-Pickup Policy — Objective & Approach

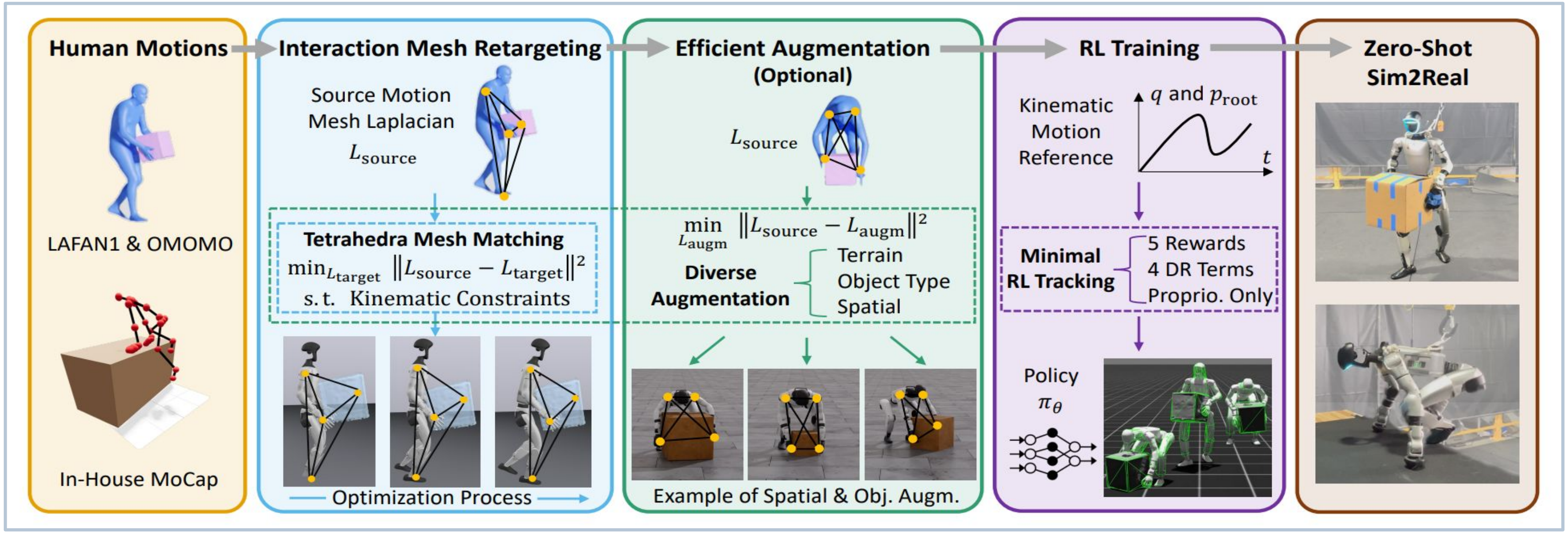
- **Task:** whole-body box pickup — walk up, two-hand grasp, lift, carry, and set down a large box.
- **Reference source:** human motion-capture clip (OmniRetarget dataset, sub3_largebox_003, SMPL-H 52-joint model).
- **Embodiment switch:** motion authored on a human SMPL-H skeleton is retargeted onto the AgiBot X2 — 31 DOF (6+6 legs, 3 waist, 7+7 arms, 2 head), 1.4 m tall, rigid half-sphere palms (no fingers).
- **Status:** policy is currently in training; deployment on the physical X2 is planned once this run completes.

Pipeline:

Human SMPL-H MoCap → Interaction-Mesh Retargeting → X2 31-DOF Reference Traj. → RL Tracking Policy (Isaac Sim / PPO) → Deploy on physical X2

Blind whole-body policy (no cameras); object state used only by the critic during training.

Retargeting Logic



Per-frame QP objective

$$\min_{\Delta q} \|L(q) - L_{human}\|^2 + w_s\|\Delta q\|^2 \quad s.t. \quad \|\Delta q\| \leq \delta, \quad q^- \leq q \leq q^+$$

$L(\cdot)$: interaction-mesh Laplacian · δ: trust-region step · $q\pm$: joint limits

Humanoid Week #1 Updates

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Past Week Objectives:

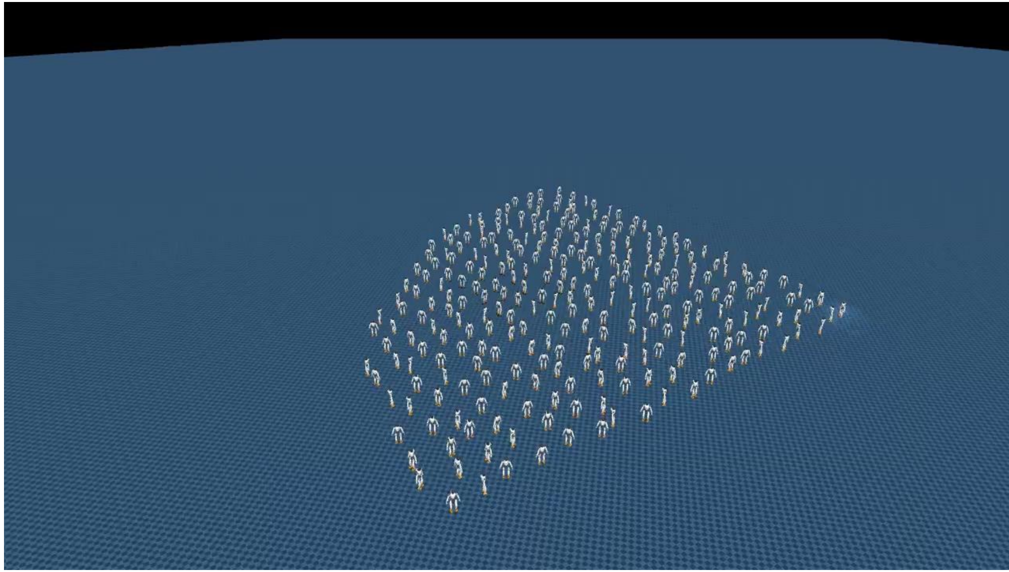
- Learn a velocity conditioned walking controller for the 31 DoF humanoid.
- Track a commanded forward, lateral, and yaw velocity while keeping the base stable.
- Train end to end in simulation with reinforcement learning, then transfer to hardware.
- The gait emerges from the reward objective. There are no reference trajectories or scripted state machines.
- Setup VLM on humanoid and run it live (In progress)

The control policy:

- A feedforward network mapping proprioceptive state to joint position targets at 50 Hz.
- **Observation:**
  - Base angular velocity and projected gravity.
  - Joint positions and velocities, and the previous action.
  - The velocity command driving the gait.
- **Action:**
  - A position target per joint, tracked by the actuator PD loops.
- Actor and critic share the observation. The critic additionally sees base linear velocity as a privileged, training only signal.

Simulation and training setup

- Training runs in MuJoCo Warp through mjlab, entirely on a single GPU.
- Roughly 4096 environments step in parallel, each an independent robot on its own terrain.
- One shared actor and critic collect experience from all environments at once.
- This yields billions of simulated timesteps within a few hours of wall clock training.



A subset of the parallel environments stepping simultaneously.

Reward design

Reward term	Purpose
Linear and angular velocity tracking	Match the commanded base velocity
Base orientation and height	Penalize torso tilt and unwanted vertical motion
Foot clearance and air time	Enforce distinct swing and stance phases
Action rate and torque	Regularize toward smooth, efficient motion
Foot slip and contact impact	Penalize sliding and hard landings
Termination on fall	End and penalize the episode when the base drops

Learning formulation

Stochastic policy over joint targets

$$a_t \sim \pi_\theta(\,\cdot \mid s_t\,)$$

Actions define PD position setpoints

$$q^{\text{target}} = q^{\text{default}} + \alpha \odot a_t$$

Reward is a weighted sum of shaped terms

$$r_t = \sum_k w_k\, r_k(s_t, a_t)$$

Velocity tracking via an exponential kernel

$$r_{\text{vel}} = \exp\left(-\frac{\|v^* - v\|^2}{\sigma^2}\right)$$

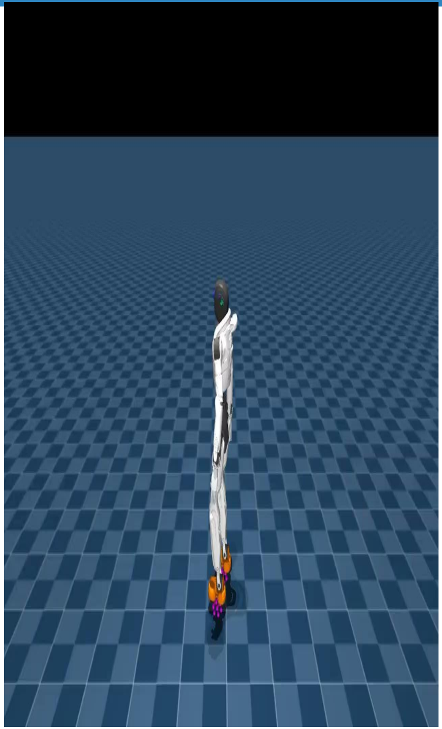
PPO maximizes the clipped surrogate

$$L(\theta) = \mathbb{E}\left[\min(\rho_t \hat{A}_t,\ \mathrm{clip}(\rho_t, 1-\varepsilon, 1+\varepsilon)\,\hat{A}_t)\right]$$

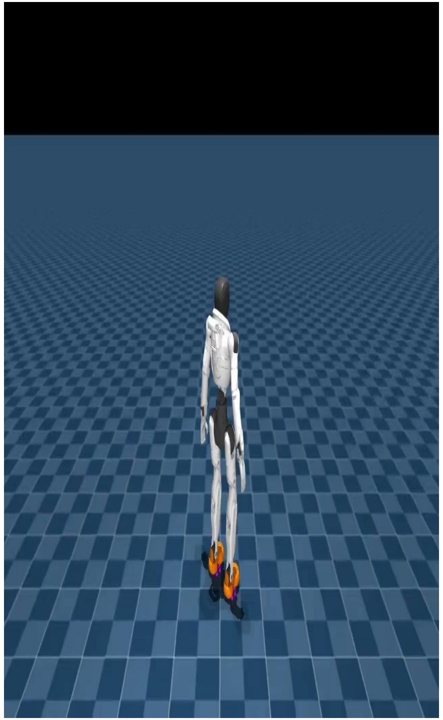
On policy optimization: rollouts are batched across all environments, advantages estimated with GAE, and the policy updated by PPO with a clipped surrogate and entropy regularization.

Policy progression over training

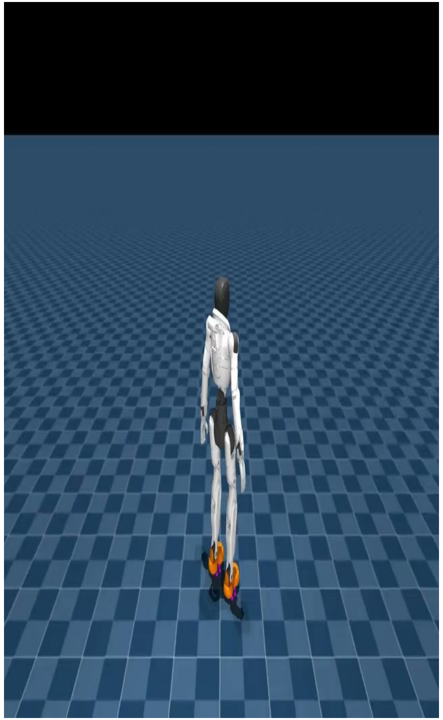
Iteration 0
Untrained, collapses immediately



Iteration 100
Unstable, fails to hold balance



Iteration 6000
Consistent locomotion

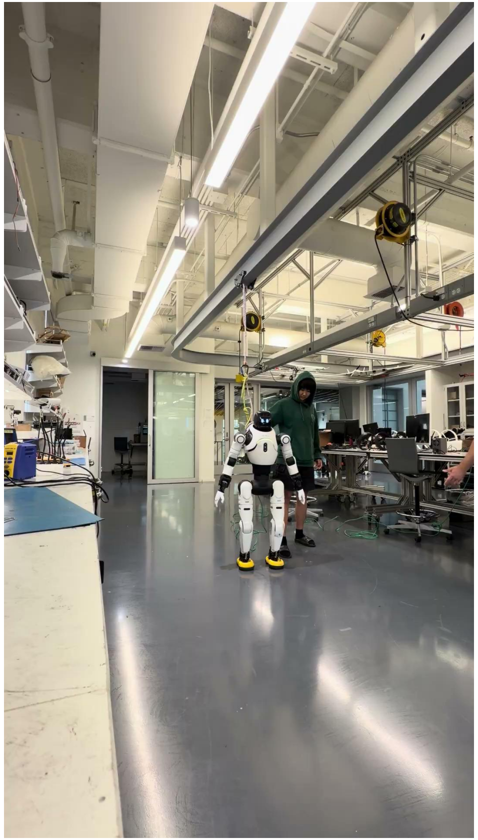


Iteration 10000
Stable, converged gait

Deterministic rollouts of saved checkpoints under a fixed forward command.

Deployment on hardware

- The trained actor is exported to a compact model and runs with numpy only on the robot.
- The deployment policy drops base linear velocity from its observation, since it is not measurable on hardware. The critic used it only in training.
- Inference runs at 50 Hz over ROS 2, and the onboard PD loops track the joint targets.
- The velocity command is streamed live, so the same policy walks in any direction on the real robot.



Policy running on the physical X2.

VLM Setup:

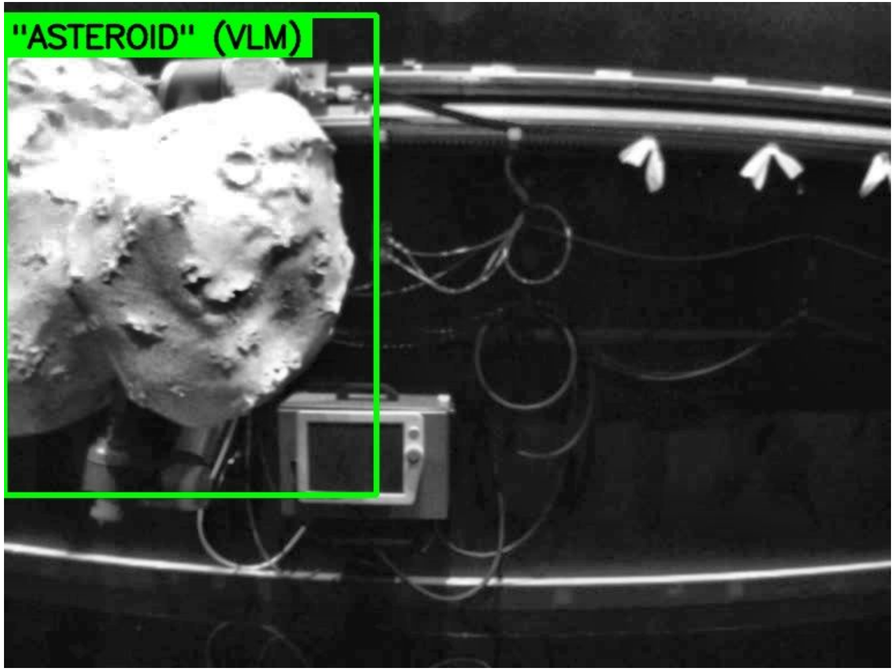
- Model: qwen2.5vl:7b via Ollama 14 frames per clip (1 every 2s, 56s video) prompted to report objects, apparatus, and any faults/damage.
- Tested the model to get performance metrics

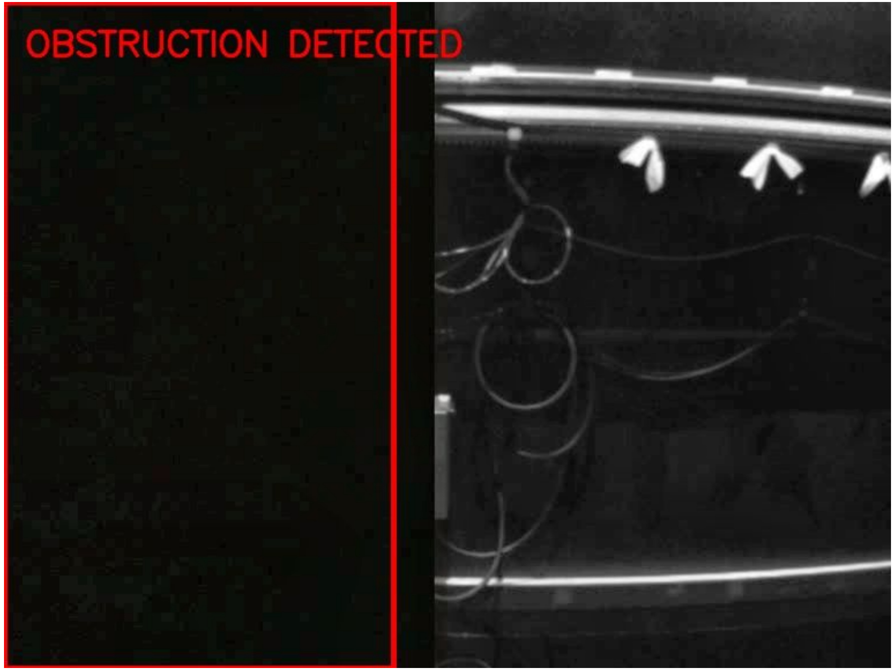
Model	VRAM	Cold load	Warm/frame	Rate	gen tok/s	Quality
qwen2.5vl:3b	2.9 GB	~6–7 s	~1.3–1.5 s	~0.7 Hz	~110	good
qwen2.5vl:7b	5.5 GB	~var	~3.8 s	~0.26 Hz	~59	best
moondream	~1.7 GB	~13 s	~2.7 s	~0.37 Hz	basic	

VLM results:

- Nominal case:
    - The image shows a robotic platform interacting with an asteroid, likely as part of a space mission to collect samples or study the object's surface properties. The robotic arm is equipped with various sensors and tools, including a camera mounted on its end, which suggests it is designed for detailed exploration and analysis.
- Faulty Case:
    - The image appears to be a split-screen view from a laboratory or engineering camera recording. On the left side of the screen, there is a predominantly dark area with minimal visible detail. The right side shows what seems to be part of an apparatus, possibly a piece of machinery or equipment, with some faint horizontal lines that might represent cables or structural elements. There are no clear indications of motion blur, faults, damage, smoke, or other unusual events in the image. The lighting is low and focused on specific areas within the right section, suggesting it may be part of an inspection or monitoring process where certain components are being highlighted for analysis.

VLM Setup:





Thanks